

Narendra Pratap Singh Rawal

Roll No.: 2023UEC1173 — Electronics and Communication Engineering

Malaviya National Institute Of Technology, Jaipur

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Education

Malaviya National Institute of Technology, Jaipur
B.Tech, Electronics and Communication Engineering

2023 – 2027
CGPA: 7.44

Swami Vivekanand Govt. Model School, Amet (Dist. Rajsamand)
Class XII, CBSE

2022
Percentage: 86.4%

Personal Projects

Research Experience: Designing of Thin Flexible Circularly Polarized Antenna for S-Band Wearable Applications

Sept 2025 – Jan 2026

- Enhanced antenna gain up to 7.3 dBi using an Artificial Magnetic Conductor (AMC) reflector.
- Performed SAR and bending analysis to validate wearable compatibility; integrated with Tx/Rx module for full system-level verification.
- **Tools & Technologies:** ANSYS HFSS, Proteus

RFID-Based Smart Toll Collection System (FASTag Inspired)

Jan 2026 – Present

- Developed an RFID-based automated toll system using the RC522 RFID module and Arduino Uno.
- Achieved wireless vehicle identification with real-time balance deduction; gained hands-on experience in SPI interfacing and embedded C programming.
- **Components & Software:** RC522 RFID Module, Arduino Uno, Arduino IDE, Proteus

IoT Sensor Monitoring Web Dashboard

Aug 2025 – Nov 2025

- Built a full-stack real-time dashboard to visualize Arduino sensor data (temperature, humidity, voltage) via a browser interface.
- Developed REST APIs with Node.js and Express.js; stored time-series readings in MongoDB with live graph updates via WebSocket.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, WebSocket, Chart.js

College Event Management Portal

Feb 2026 – Present

- Developed a MERN stack web application for managing college fests, event registrations, and announcements.
- Implemented JWT-based authentication, role-based access control (Admin/Student), and automated email notifications using Nodemailer.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS, Vercel

Technical Skills and Interests

Simulation Tools: MATLAB, LTspice, ANSYS HFSS, Proteus, Cadence

Programming & Web: C/C++, Python, JavaScript, React.js, Node.js, Express.js, MongoDB, HTML/CSS, Tailwind CSS

Dev Tools: Git, GitHub, VS Code, Arduino IDE, Postman, Vercel, Render

Coursework: Microprocessors (8085 & 8086), Digital Design, VLSI Testing and Testability, Antenna Theory, Satellite Communication, DSA

Areas of Interest: Antenna Design, Satellite Communication, Full-Stack Web Development, Embedded Systems